

$\dim(V) = n$ , Vector space  $V$  over  $\mathbb{R}$ ,  $B$  Basis

Linear transformation

$$T: V \rightarrow V$$

Matrix  $M_B(T)$   
 $\downarrow$   
 $n \times n$  matrix

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$T(X) = AX$$

'A' be an  $n \times n$  matrix

where  $X$  is column vector in  $\mathbb{R}^n$

•  $A$  is matrix associated with w.r.t st. basis of  $\mathbb{R}^n$

•  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$   
 $B \rightarrow$  st basis of  $\mathbb{R}^3$

$$M_B(T) = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$T(x, y, z) = (x+y, y+z, z)$$

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x+y \\ y+z \\ z \end{pmatrix}$$

• Eigen vector :-  $T: V \rightarrow V$ ,  $\wedge 0 \neq v \in V$  is called Eigen vector of  $T$  if  $T(v) = \alpha v$  for some  $\alpha \in F$ . ' $\alpha$ ' is called eigen value corresponding to Eigen vector  $v$ .

\* A linear transformation  $T: V \rightarrow V$  is diagonalizable iff  $\exists$  a Basis  $B$  of  $V$  consisting of

Eigen vectors of T.

• Eigen vector and Eigen value of a matrix

Diagonalizable Matrix :- An  $n \times n$  matrix  $A$  is called diagonalizable if it is similar to a diagonal matrix.

\* If  $\exists$  an invertible matrix  $P$  s.t.  $P^{-1}AP$  is a diagonal matrix.

• Let  $A$  be diagonalizable matrix

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$T(x) = AX$$

$\rightarrow$   $A$  is matrix associated with  $T$  w.r.t standard basis.

If  $T$  is diagonalizable, then  $\exists$  basis  $B$  of  $\mathbb{R}^n$  s.t.  $M_B(T) = \text{Diagonal matrix}$

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

s.t. Basis  $\swarrow$   $A$   $\searrow$   $B$

$$M_B(T) = \text{Diagonal matrix}$$

$$A = P^{-1} M_B(T) P \Rightarrow A \text{ is diagonal matrix}$$

\* Eigen vector :- Let  $A$  be an  $n \times n$  matrix.  $A$  non zero column vector  $X$  in  $\mathbb{R}^n$  is called Eigen vector of  $A$  if  $AX = \alpha X$ , for some  $\alpha \in \mathbb{R}$  and ' $\alpha$ ' is called eigen value of matrix  $A$ .

$$T: \mathbb{R}^n \rightarrow \mathbb{R}^n$$

$$T(x) = \alpha X$$

$$\Rightarrow T(x) = \alpha X$$

vector of  $T$ .

$$T: \mathbb{R} \rightarrow \mathbb{R}$$

$$T(x) = Ax = \alpha x$$

$$\Rightarrow T(x) = \alpha x$$

$\Rightarrow x$  is eigen vector of  $T$ .

Question:- find Eigen values and Eigen vectors of  $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$  defined by

$$T(x, y) = (2x, x+y)$$

(Decide whether  $T$  is diagonalizable or not?)

Solution:-  $(0,0) \neq (x,y)$ ,  $T(x,y) = \alpha(x,y)$   
 $(2x, x+y) = (\alpha x, \alpha y)$

$$\begin{array}{l|l} \begin{array}{l} 2x = \alpha x \\ 2x - \alpha x = 0 \\ (2 - \alpha)x = 0 \end{array} & \begin{array}{l} \alpha = 2 \\ x = 0 \end{array} \\ \hline \begin{array}{l} x + y = \alpha y \\ x + y = 2y \Rightarrow x = y \\ y = \alpha y \Rightarrow (1 - \alpha)y = 0 \end{array} & \begin{array}{l} \alpha = 1 \\ y = 0 \end{array} \end{array}$$

$$\boxed{\alpha = 2, x = y}$$

$$\alpha = 2, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\boxed{\alpha = 1, x = 0}$$

$$\alpha = 1, \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$T(1,1) = (2,2) = 2(1,1)$$

$$T(x,x) = (2x,2x) = 2(x,x)$$

$$T(0,1) = (0,1) = 1(0,1)$$

•  $\{(1,1), (0,1)\}$  a Basis of  $\mathbb{R}^2$ ?? Yes  
 $T$  is diagonalizable.

$$T(1,1) = (2,2) = 2(1,1) + 0(0,1)$$

$$T(0,1) = (0,1) = 0(1,1) + 1(0,1)$$

$$\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Question:- find Eigen values and Eigen vectors of linear transformation  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  defined by

if  $T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$  defined by

$$T(x, y, z) = (x, x+y, y-z).$$

Decide if  $T$  is diagonalizable or not??

Solution:-  $(0,0,0) \neq (x,y,z)$  s.t.

$$T(x, y, z) = \alpha(x, y, z)$$

$$(x, x+y, y-z) = (\alpha x, \alpha y, \alpha z)$$

$$\begin{aligned} x &= \alpha x \\ x - \alpha x &= 0 \\ (1-\alpha)x &= 0 \\ \alpha &= 1 \quad \downarrow \quad x=0 \end{aligned}$$

$$\left| \begin{array}{l} x+y = \alpha y \\ \boxed{\alpha=1} \quad x+y=y \Rightarrow \underline{x=0} \\ \boxed{x=0} \quad y = \alpha y \\ y - \alpha y = 0 \rightarrow y=0 \end{array} \right| \begin{array}{l} y-z = \alpha z \\ \boxed{\alpha=1}, \quad y-z=z \\ y=2z \\ \boxed{y=0} \quad -z = \alpha z \\ \alpha z + z = 0 \\ (\alpha+1)z = 0 \\ \alpha = -1 \quad \downarrow \quad z=0 \end{array}$$

$$\boxed{\alpha=1, x=0, y=2z} \rightarrow [0, 2, 1]$$

$$\boxed{\alpha=-1, x=0, y=0} \rightarrow [0, 0, 1]$$

$$T(0, 2, 1) = (0, 2, 1) = 1(0, 2, 1)$$

$$T(0, 0, 1) = (0, 0, -1) = -1(0, 0, 1)$$

$$\{ [0, 2, 1], [0, 0, 1] \}$$

is L.I. set.  
Not a Basis

T is Not Diagonalizable

Let  $T: V \rightarrow V$  be linear transformation, then Eigen vectors corresponding to distinct eigen values are linearly independent vectors.

$T: V \rightarrow V$  (linear transformation)

are linearly

$$T: V \rightarrow V$$

(linear transformation)  
dim  $V = n$



\*  $\{v_1, v_2, \dots, v_n\}$  is linearly independent set.

◦ If  $\dim V = n$ ,  $T: V \rightarrow V$ , then  $T$  can have at most  $n$  distinct eigen values.

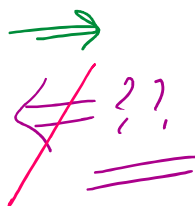
$\therefore$  Any subset of  $V$  containing more than  $n$  vectors will be linearly dependent.

Let  $\dim V = n$ .

$T: V \rightarrow V$  is a L.T.

$T$  has  $n$  distinct eigen values

$T$  is diagonalizable.



$$\begin{matrix} \lambda_1 & \lambda_2 & \dots & \lambda_n \\ | & | & & | \\ v_1 & v_2 & & v_n \end{matrix}$$

$\{v_1, v_2, \dots, v_n\}$  is L.I. set

$\hookrightarrow$  Basis of  $V$

Example:-

$$T: \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

$$T(x, y, z) = (2x, 2y, 2z)$$

$$B = \{ (1, 0, 0), (0, 1, 0), (0, 0, 1) \}$$

$$M_{\mathbb{Q}}(T) = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

$$\underline{T(1,0,0) = (2,0,0)}, \quad \underline{T(0,1,0) = (0,2,0)} \quad \underline{T(0,0,1) = (0,0,2)}$$

• Find Eigen values and Eigen vectors of  $T$ .

$$(x,y,z) \neq (0,0,0) \quad T(x,y,z) = \alpha(x,y,z)$$

$$(2x, 2y, 2z) = (\alpha x, \alpha y, \alpha z)$$

$$\begin{array}{l} 2x = \alpha x \\ \swarrow \quad \searrow \\ \alpha = 2 \quad x = 0 \end{array}$$

$$\boxed{\alpha = 2, (1, 0, 0)}$$

$$\begin{array}{l} 2y = \alpha y \\ \swarrow \quad \searrow \\ \alpha = 2 \quad y = 0 \end{array}$$

$$\boxed{\alpha = 2, (0, 1, 0)}$$

$$\begin{array}{l} 2z = \alpha z \\ \swarrow \quad \searrow \\ \alpha = 2 \quad z = 0 \end{array}$$

$$\boxed{\alpha = 2, (0, 0, 1)}$$

$\{(1,0,0), (0,1,0), (0,0,1)\}$  is Basis of  $\mathbb{R}^3$

• Let  $\dim V = n$ ,  $T: V \rightarrow V$  is diagonalizable iff  
 $T$  has  $n$  L.I. eigen vectors.

However,  $T$  may not have  $n$  distinct eigen values